Exam AWS Certified Developer - Associate DVA-C02 All Questions

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EXAM AWS CERTIFIED DEVELOPER - ASSOCIATE DVA-C02 TOPIC 1 QUESTIO...

Exam question from Amazon's AWS Certified Developer - Associate DVA-C02
Question #: 128
Topic #: 1
[All AWS Certified Developer - Associate DVA-C02 Questions]

A company has a front-end application that runs on four Amazon EC2 instances behind an Elastic Load Balancer (ELB) in a production environment that is provisioned by AWS Elastic Beanstalk. A developer needs to deploy and test new application code while updating the Elastic Beanstalk platform from the current version to a newer version of Node.js. The solution must result in zero downtime for the application.

Which solution meets these requirements?

A. Clone the production environment to a different platform version. Deploy the new application code, and test it. Swap the environment URLs upon verification.

B. Deploy the new application code in an all-at-once deployment to the existing EC2 instances. Test the code. Redeploy the previous code if verification fails.

C. Perform an immutable update to deploy the new application code to new EC2 instances. Serve traffic to the new instances after they pass health checks.

D. Use a rolling deployment for the new application code. Apply the code to a subset of EC2 instances until the tests pass. Redeploy the previous code if the tests fail.

Show Suggested Answer

by MrTee at *April 23, 2023, 12:45 a.m.*

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Comments

MrTee Highly Voted 2 years, 11 months ago

Selected Answer: C

Option C is the correct solution that meets the requirements. Performing an immutable update to deploy the new application code to new EC2 instances and serving traffic to the new instances after they pass health checks will ensure zero downtime for the application.

Option A would work but cloning the production environment to a different platform version will result in a longer deployment time and can impact the cost of the environment.

upvoted 23 times

awsdummie 2 years, 11 months ago

C is incorrect, after passing health checks the elastic Beanstalk transfers them to the original Auto Scaling group. No testing or platform update is done.

upvoted 5 times

yeacuz 2 years, 10 months ago

I would agree that option A can affect the cost, but cost is not the issue. The question is asking for zero downtime. I believe the answer is option A

upvoted 3 times

gagol14 Highly Voted 2 years, 9 months ago

Selected Answer: A

Not C: While an immutable update can ensure zero downtime during the deployment process, it doesn't account for updating the Elastic Beanstalk platform version.

upvoted 11 times

Artemiy Most Recent 1 year ago

Selected Answer: A

Option A creates a completely separate environment with the newer Node.js version, deploys the new code there, and allows for testing without affecting the production environment. Once verified, you simply swap the URLs

upvoted 1 times

sumanshu 1 year, 3 months ago

Selected Answer: C

A) Eliminated - This solution meets the requirement of zero downtime but may have additional resource and cost overhead.

B) Eliminated - All-at-once deployment means that all instances are updated at the same time, which would result in downtime during the deployment.

C) Correct - Immutable updates ensure that new EC2 instances are created, and only healthy instances will serve traffic. The old EC2 instances are not affected until the new instances are confirmed to be healthy and live.

D) Eliminated - Rolling deployments help ensure that some EC2 instances are always running the old code and serving traffic while others are being updated.

upvoted 1 times

MasoudK 1 year, 6 months ago

Option A is Correct not C: By cloning the production environment to a different platform version, you create a separate environment where you can safely deploy and test the new application code and platform version without affecting the live production environment.• Option C (Immutable update): While immutable updates ensure zero downtime by deploying to new instances, they do not address the need to update the Elastic Beanstalk platform version. Additionally, this approach can be more resource-intensive and costly.

upvoted 1 times

Saudis 1 year, 6 months ago

Selected Answer: C

Ans is c

upvoted 1 times

65703c1 1 year, 10 months ago

Selected Answer: C

C is the correct answer.

upvoted 1 times

ibratoev 2 years ago

It is A:
https://docs.amazonaws.cn/en_us/elasticbeanstalk/latest/dg/using-features.platform.upgrade.html#using-features.platform.upgrade.config

upvoted 3 times

KarBiswa 2 years ago

This question must be true for 2 options because C & D are both correct

upvoted 1 times

KarBiswa 2 years ago

Selected Answer: D

https://docs.amazonaws.cn/en_us/elasticbeanstalk/latest/dg/using-features.rolling-version-deploy.html

upvoted 1 times

SerialiDr 2 years, 2 months ago

Selected Answer: C

The solutions that best meet the requirements for zero downtime are:

A. Clone the production environment to a different platform version. Deploy the new application code, and test it. Swap the environment URLs upon verification.
C. Perform an immutable update to deploy the new application code to

new EC2 instances. Serve traffic to the new instances after they pass health checks.
Both options A and C provide robust strategies for deploying updates with zero downtime, allowing for thorough testing in an isolated environment before directing production traffic to the new setup.

upvoted 3 times

Certified101 2 years, 3 months ago

Selected Answer: A

Not C: tt doesn't account for updating the Elastic Beanstalk platform version. This would affect both the live and test environments.

Its also best practise to have 2 seperate environments for production and test and there is no mention of cost optimisation here.

upvoted 3 times

tqiu654 2 years, 4 months ago

Selected Answer: A

ChatGPT:A

upvoted 2 times

Rameez1 2 years, 5 months ago

Selected Answer: C

A & C both works for given scenario but C does it more feasibly for Elastic Beanstalk with zero downtime.

upvoted 1 times

stilloneway 2 years, 7 months ago

Selected Answer: C

Key terminology in question is "Test". So it should be immutable for quick rollback in case of test not working.

upvoted 2 times

CrescentShared 2 years, 2 months ago

It's a downtime if test fails and rollback.

upvoted 1 times

love777 2 years, 7 months ago

Selected Answer: C

Explanation:

Immutable Update with Elastic Beanstalk:
With an immutable update, Elastic Beanstalk provisions new instances with the updated code while keeping the existing instances running. The traffic is shifted gradually to the new instances after they pass health checks, ensuring that there is no downtime during the deployment. If any issue arises during the deployment, traffic is still being served by the existing instances.

upvoted 4 times

Naj_64 2 years, 7 months ago

Selected Answer: D

Screenshot of Step 4 of Method 1 in the link:
https://docs.amazonaws.cn/en_us/elasticbeanstalk/latest/dg/using-features.platform.upgrade.html#using-features.platform.upgrade.config

"...your application is unavailable during the update. To keep at least one instance in service during the update, enable rolling updates"

upvoted 2 times

Naj_64 2 years, 7 months ago

I take this back. I'm going with A

"However, you can avoid this downtime by deploying the new version to a separate environment. The existing environment's configuration is copied and used to launch the green environment with the new version of the application. The new green environment will have its own URL. When it's time to promote the green environment to serve production traffic, you can use Elastic Beanstalk's Swap Environment URLs feature."

https://docs.aws.amazon.com/whitepapers/latest/blue-green-deployments/swap-the-environment-of-an-elastic-beanstalk-application.html

upvoted 2 times